Theory question on Diffraction

1. What is diffraction of light? Write the condition of maxima and minima due to Fraunhofer diffraction at a double slit with a neat ray diagram.

2. What is a diffraction grating? Explain the construction and working of a plane transmission diffraction grating. Define the grating element and write the grating equation.

3. Explain how the wavelength of monochromatic light can be determined using a diffraction grating.

4. What is meant by the resolving power of an optical instrument? State and explain Rayleigh's criterion of resolution.

5. Define the resolving power of a diffraction grating. Write the expression for resolving power of a diffraction grating with meaning of symbols used. What are the factors affecting the resolving power of a grating.

6. What are formulae for resolving power of a diffraction grating? Write relation between these formulae for well resolved, just resolved and not resolved.